

Erdős Problem #408

EP-408

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\phi(n)$ be the Euler totient function and $\phi_k(n)$ be the iterated ϕ function, so that $\phi_1(n) = \phi(n)$ and $\phi_k(n) = \phi(\phi_{k-1}(n))$. Let $f(n) = \min\{k : \phi_k(n) = 1\}$. Does $f(n)/\log n$ have a distribution function? Is $f(n)/\log n$ almost always constant? What can be said about the largest prime factor of $\phi_k(n)$ when, say, $k = \log \log n$?

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>